

Notes: Kacperczyk and Schnabl (QJE, 2013)

How Safe Are Money Market Funds?

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1 Big picture

- **Question.** How safe are money market funds? The paper studies whether funds that look cash-like actually have incentives to take risk and whether this risk creates run vulnerability.
- **Setting.** The setting is the U.S. money market fund industry during the financial crisis of 2007-2010, with special emphasis on institutional prime money market funds.
- **Core institutional feature.** Money market funds are not insured bank deposits, but they try to maintain a stable net asset value of \$1 per share. This makes them look safe to investors while leaving them exposed to asset losses and runs.
- **Main shock.** Starting in August 2007, the yields on risky money market instruments rose sharply relative to Treasury and agency debt. Funds suddenly had a meaningful choice between safer assets and higher-yield, riskier assets.
- **Main incentive.** Fund inflows are highly sensitive to yields. A fund that raises current yield attracts assets, and sponsors earn fees as a fraction of assets under management.
- **Main cross-sectional mechanism.** Sponsors with little non-money-fund business have less franchise value to protect, so they are more willing to take risk. Sponsors with large non-money-fund businesses or broader financial conglomerate reputations internalize larger spillover costs if a money fund fails.
- **Main empirical result.** Funds sponsored by firms with more money fund business, or by stand-alone asset managers, take more risk after August 2007. Funds sponsored by firms with larger non-money-fund businesses take less risk.
- **Run result.** Risk taking matters ex post: funds that took more risk before Lehman's default experienced larger redemptions. Sponsors with broader businesses were more likely to support funds and/or later reshape their money fund business.
- **Economic takeaway.** Money market funds are not mechanically safe cash equivalents. Their safe appearance can coexist with strong incentives to chase yield, and that incentive can make short-term funding markets more fragile.

2 Incremental contribution of the paper

- **From money funds as safe cash to money funds as risk-taking intermediaries.** The paper shows that the money market fund sector can behave like a risk-taking financial intermediary when risk premia open up.
- **A crisis-based opportunity shock.** The repricing of money market instruments after August 2007 creates a sharp change in investment opportunities. Before the crisis, safe and risky eligible instruments had similar yields; after the crisis, risky instruments paid much higher spreads.
- **A direct incentive test.** The paper documents a strong flow-performance relationship among institutional prime funds: investors chase yield, so fund sponsors can gain assets and fees by increasing yields.

- **A sponsor-franchise-value contribution.** The paper links risk choices to sponsor business structure. Sponsors with more non-money-fund business have more to lose from reputational spillovers and therefore take less money fund risk.
- **A shadow-banking contribution.** The paper studies a deposit-like, uninsured, non-bank intermediary and shows how runnable short-term claims can encourage rather than discipline risk taking.
- **A run-consequences contribution.** The paper connects ex ante risk taking to ex post redemptions, sponsor support, exits, and fund name changes after Lehman’s default.
- **A regulatory contribution.** The findings speak directly to the design of money market fund regulation: restrictions on asset eligibility and stable NAV accounting do not eliminate incentives to reach for yield.
- **A data contribution.** The authors assemble a weekly fund-level data set with yields, flows, holdings by instrument type, maturities, sponsor characteristics, ratings, and post-Lehman support events.

3 Data and measurement

3.1 Money market fund data

- The primary data source is iMoneyNet, which covers the universe of money market funds.
- The sample period is January 2005 to September 2011, though the main risk-taking tests focus on January 2006 to August 2008.
- iMoneyNet provides weekly fund-level information on yields, expense ratios, total assets, average maturity, holdings by instrument type, and fund sponsor.
- The analysis focuses on taxable prime funds, especially institutional prime funds, because these funds can invest in private money market instruments and institutional investors are especially yield sensitive.
- Treasury funds are not the main object of study because their portfolios are limited to government-related securities and therefore are less exposed to the private-credit risk shock.

Interpretation: The data are unusually high frequency and contain portfolio composition. This lets the paper observe actual risk-taking behavior, not just returns or flows.

3.2 Fund and sponsor variables

- **Spread** is the annualized gross fund yield minus the one-month Treasury bill rate.
- **Fund Flow** is the percentage change in total net assets from week t to $t + 1$, adjusted for earned interest and winsorized.
- **Holdings Risk** is based on the share of riskier assets, especially bank obligations net of safer holdings such as repurchase agreements and Treasuries.

- **Maturity Risk** is the weighted average maturity of the fund portfolio. Longer maturity is riskier because it raises exposure to interest-rate and liquidity risk.
- **Fund Business** is the sponsor's mutual fund assets other than institutional prime money fund assets divided by the sponsor's total mutual fund assets. A high value means the sponsor has more non-money-fund franchise value to protect.
- **Conglomerate** equals one if the sponsor is affiliated with a commercial bank, investment bank, or insurance company, and zero for stand-alone asset managers.

Interpretation: The key empirical distinction is between sponsors that are specialized in money funds and sponsors that have broader businesses. The latter face larger reputational spillover costs from a money fund failure.

3.3 Risk-taking opportunity measures

- The paper classifies instruments into safer and riskier categories.
- Safer instruments include Treasuries, agency debt, and repurchase agreements.
- Riskier instruments include bank obligations, floating-rate notes, commercial paper, and asset-backed commercial paper.
- The key opportunity shock is the post-August-2007 widening of spreads on risky instruments relative to safer government-related instruments.

Interpretation: Before August 2007, regulation and low spreads left little room for funds to differentiate themselves by risk. After August 2007, the same regulatory menu allowed materially different risk choices.

3.4 Post-Lehman outcome variables

- **Redemptions** measure fund outflows in the week after Lehman's bankruptcy.
- **Support** is an indicator for whether the sponsor filed a no-action letter with the SEC to provide support.
- **Exit** indicates whether the fund closed after the government guarantee expired.
- **Name change** captures whether the fund changed its name to more closely match the sponsor's name, consistent with making sponsor backing more salient.

Interpretation: These variables let the paper connect ex ante risk choices to ex post crisis consequences.

4 Models and empirical specifications

4.1 Instrument return regression

The first empirical step estimates how different instrument holdings translate into fund yield spreads:

$$Spread_{i,t+1} = \alpha_i + \gamma_t + \sum_j \beta_j Holdings_{i,j,t} + \delta' X_{i,t} + \varepsilon_{i,t+1}.$$

- $Spread_{i,t+1}$ is fund i 's gross yield spread over the Treasury bill rate.
- $Holdings_{i,j,t}$ is the share of fund assets invested in instrument type j .
- The omitted category is Treasuries and agency debt.
- Fund and week fixed effects absorb persistent fund characteristics and aggregate time effects.
- The coefficients β_j measure the yield premium on each instrument type relative to Treasuries and agency debt.

Interpretation: This model quantifies the risk-return opportunity set. If risky instruments pay little extra yield, funds have little incentive to reach for yield. If they pay much more, risk taking becomes attractive.

4.2 Flow-performance regression

The second step estimates whether investors chase yield:

$$FundFlow_{i,t+1} = \gamma_t + \beta Spread_{i,t} + \delta' X_{i,t} + \varepsilon_{i,t+1}.$$

- The dependent variable is next week's fund flow.
- The key independent variable is the current fund spread.
- Controls include fund size, expenses, age, flow volatility, and family size.
- Some specifications include fund fixed effects.
- The paper also interacts spread with sponsor characteristics to test whether investors risk-adjust yields using sponsor type.

Interpretation: A positive β means high-yield funds attract assets. Because sponsors earn fees on assets under management, this creates a direct incentive to raise yields by holding riskier assets.

4.3 Business-spillovers and risk-taking regression

The main risk-taking design is a difference-in-differences specification:

$$Risk_{i,t+1} = \alpha + \gamma_t + \beta_1 BusinessSpillovers_{i,2006} + \beta_2 Post_t \times BusinessSpillovers_{i,2006} + \delta' X_{i,2006} + \varepsilon_{i,t+1}.$$

- *Risk* is measured using holdings risk, maturity risk, or spread.
- *BusinessSpillovers* is either Fund Business or Conglomerate.
- *Post* equals one from August 2007 to August 2008.
- The controls are measured before the crisis, in January 2006.
- Standard errors are clustered at the sponsor level.

Interpretation: Since Fund Business is the share of non-institutional-prime-fund business, a negative post interaction means that sponsors with more outside franchise value take less risk once risk-taking opportunities expand.

4.4 Post-Lehman regressions

The paper then studies crisis consequences using cross-sectional regressions such as:

$$Redemptions_i = \alpha + \beta_1 BusinessSpillovers_i + \beta_2 Spread_i + \delta' X_i + \varepsilon_i,$$

and similar models for support, exit, and name changes:

$$Support_i = \alpha + \beta BusinessSpillovers_i + \delta' X_i + \varepsilon_i,$$

$$Exit_i = \alpha + \beta BusinessSpillovers_i + \delta' X_i + \varepsilon_i,$$

$$NameChange_i = \alpha + \beta BusinessSpillovers_i + \delta' X_i + \varepsilon_i.$$

Interpretation: These regressions ask whether pre-run risk and sponsor structure predict what happens when the safe-money-fund perception breaks down after Lehman's default.

5 Identification and empirical design

5.1 What identifies the main effect

- The key empirical shock is the expansion in risk-taking opportunities after August 2007.
- Before August 2007, eligible money market instruments offered similar yields, so there was little payoff to changing risk.
- After August 2007, risky instruments paid much higher spreads, creating a stronger incentive to reach for yield.

- Cross-sectional variation in sponsor franchise value is measured before the crisis, reducing concerns that it was chosen in response to the opportunity shock.

Interpretation: The design is strongest when read as: once the opportunity to take risk suddenly becomes valuable, sponsors with different franchise values respond differently.

5.2 Why sponsor business spillovers matter

- A money fund failure can hurt the sponsor’s broader business through reputational spillovers.
- Sponsors with large non-money-fund businesses face potentially large losses if investors lose trust in the sponsor.
- Stand-alone asset managers and sponsors heavily specialized in money funds have less non-money-fund business to protect.
- Therefore, specialized sponsors should be more willing to raise yield by taking risk.

Interpretation: The mechanism is not simply that all sponsors like risk. The incentive depends on the sponsor’s payoff function: fee gains from inflows versus expected franchise losses if the fund fails.

5.3 Robustness and placebo logic

- **Fund and sponsor fixed effects:** The paper controls for persistent fund or sponsor differences.
- **Retail-fund placebo:** Retail investors are less yield sensitive. The same sponsor characteristics should matter less for retail funds if institutional yield chasing is the mechanism.
- **Post-government-guarantee test:** If risk taking were driven by expected government bailouts, the business-spillover pattern should persist after explicit guarantees. It largely disappears.
- **Financial-strength test:** Among stand-alone asset managers, the effect remains after controlling for sponsor credit-rating status.
- **Outlier and Reserve Primary checks:** Results are not driven only by a few extreme funds or by the Reserve Primary Fund.

Interpretation: The robustness tests are designed to separate the reputation/spillover channel from unobserved sponsor risk culture, implicit government support, and sponsor financial strength.

5.4 Limitations

- The design is not a randomized experiment; unobserved time-varying sponsor traits cannot be ruled out completely.
- Risk measures are based on instrument categories, maturity, and yield spreads; they cannot perfectly observe all issuer-level risk.
- The post-Lehman support data rely on observed support actions and no-action letters, which may not capture every informal sponsor action.

- External validity is strongest for institutional prime funds during the crisis period; Treasury funds and retail funds have different incentives.

Interpretation: The evidence is most persuasive as a crisis-based, cross-sectional test of risk-taking incentives within institutional prime money market funds.

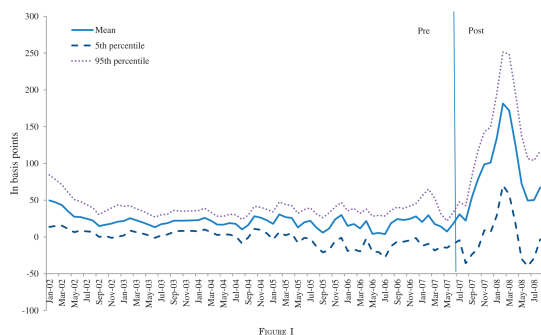
6 Tables and figures

6.1 Figure I and Figure II: Yield dispersion and instrument spreads

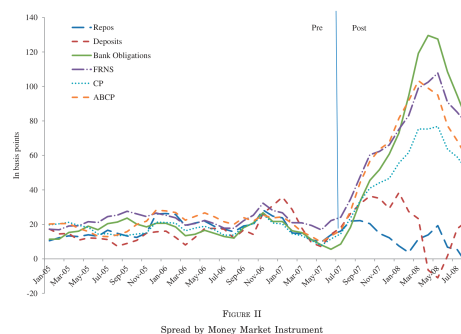
Read:

- Figure I shows that average fund yields and yield dispersion were low before August 2007.
- After August 2007, the mean spread of money fund yields over Treasury bills rises sharply and the cross-sectional dispersion widens.
- Figure II shows why this happens: risky instruments such as bank obligations, floating-rate notes, commercial paper, and ABCP begin to pay much higher spreads relative to Treasuries and agencies.
- Safer instruments such as repos and deposits do not experience the same persistent spike.

Economic message: The crisis created the central experiment of the paper. Funds suddenly had a meaningful choice: remain close to safe government-related instruments or earn higher yields by moving into riskier money market instruments.



(a) Figure I: dispersion in fund yields



(b) Figure II: spread by instrument

Figure I and Figure II: risk-taking opportunities expand after August 2007.

6.2 Figure III: Reserve Primary Fund versus Fidelity Institutional Prime

Read:

- Figure III compares the Reserve Primary Fund (RPF) and Fidelity Institutional Prime (FID).
- Before August 2007, both funds have spreads close to the industry average.

- After August 2007, RPF's spread rises sharply relative to the industry, while FID's spread remains much more stable.
- RPF also experiences much stronger asset growth, consistent with investors chasing its higher yield.

Economic message: The two-fund comparison makes the mechanism concrete. A high-yield strategy can attract assets, but it also creates exposure to risky securities that can become fatal during a run.

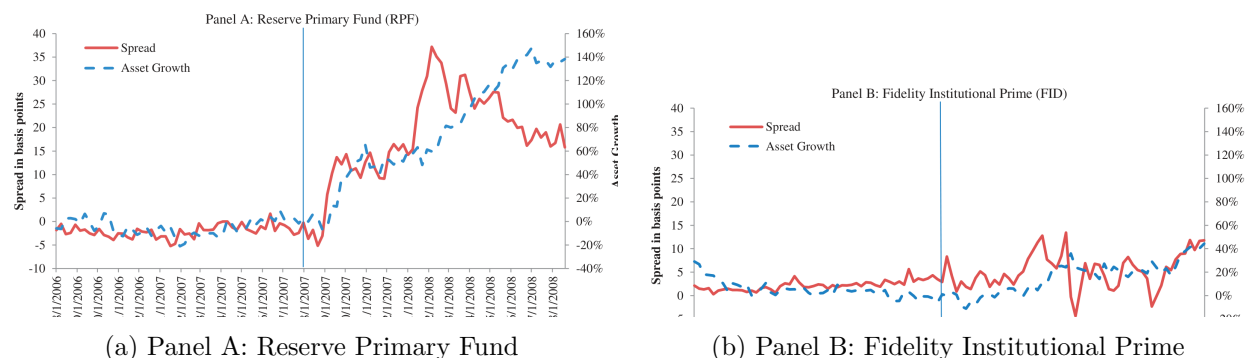


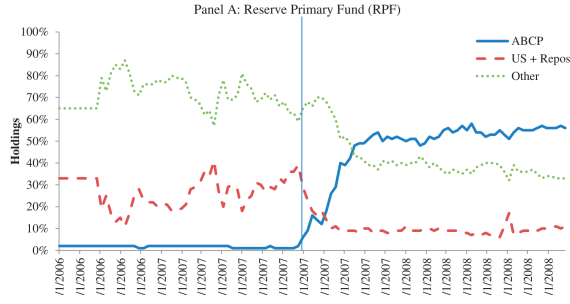
Figure III: relative performance and assets.

6.3 Figure IV: Asset holdings of Reserve Primary and Fidelity

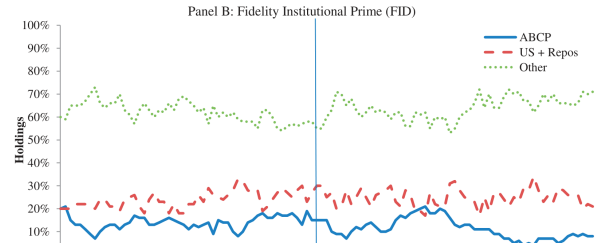
Read:

- Figure IV shows the portfolio choices behind the performance patterns in Figure III.
- RPF increases its exposure to ABCP and reduces safer U.S. securities plus repos after August 2007.
- FID's holdings remain more stable and do not show the same shift into ABCP.
- The difference is consistent with the idea that sponsors with more franchise value at stake avoid aggressive yield-chasing.

Economic message: The fund yield difference is not just accounting noise. It reflects different portfolio risk choices.



(a) Panel A: Reserve Primary Fund holdings



(b) Panel B: Fidelity holdings

Figure IV: asset holdings.

6.4 Table I: Summary statistics

Read:

- Table I summarizes 148 institutional prime money market funds as of January 2006.
- The average fund size is about \$4.9 billion, and the average spread is 6.93 basis points.
- Funds hold commercial paper, floating-rate notes, bank obligations, ABCP, repos, Treasuries/agencies, and deposits.
- The table splits funds into high and low Fund Business groups. High Fund Business means the sponsor has more non-money-fund assets to protect.
- The two groups are broadly similar on many observable characteristics, which helps the later cross-sectional design.

Economic message: Table I establishes that the key sponsor-business variable is not simply picking up obviously different fund types before the crisis.

TABLE I
SUMMARY STATISTICS OF INSTITUTIONAL PRIME MONEY MARKET FUNDS

	(1) All	(2) High FB	(3) Low FB
Fund characteristics			
Spread (bp)	6.93 (6.44)	6.60 (7.54)	7.28 (5.00)
Expense ratio (bp)	31.64 (19.10)	32.40 (18.43)	30.81 (19.90)
Fund size (\$mil)	4,886 (8,685)	2,981 (4,833)	6,951*** (11,169)
Maturity (days)	34.32 (11.02)	35.12 (12.48)	33.45 (9.17)
Age (years)	10.61 (4.75)	10.43 (5.53)	10.81 (3.75)
Family size (\$bil)	72.8 (149.1)	97.5 (200.9)	45.9** (39.2)
Fund business	0.764 (0.198)	0.897 (0.064)	0.619*** (0.192)
Conglomerate (in %)	60.1 (49.1)	55.8 (50.0)	64.8 (48.1)
Portfolio holdings			
U.S. Treasuries & agency	0.060 (0.109)	0.072 (0.120)	0.048 (0.095)
Repurchase agreements	0.135 (0.150)	0.142 (0.169)	0.126 (0.128)
Bank deposits	0.032 (0.057)	0.021 (0.039)	0.044** (0.069)
Bank obligations	0.122 (0.126)	0.111 (0.120)	0.135 (0.132)
Floating-rate notes	0.198 (0.162)	0.192 (0.168)	0.204 (0.156)
Commercial paper	0.320 (0.224)	0.356 (0.252)	0.280** (0.182)
Asset-backed CP	0.134 (0.155)	0.106 (0.151)	0.164** (0.154)
Funds	148	77	71

Notes. This table shows summary statistics for all U.S. institutional prime money market funds as of January 1, 2006. *Fund Business (FB)* is mutual fund assets other than institutional prime fund assets as a share of total sponsor's mutual fund assets. High (Low) FB includes all funds with *Fund Business* above (below) the median value of *Fund Business* (81.6%). Fund characteristics are spread, expenses, fund size, average portfolio maturity, age, family size (mutual funds assets other than money market fund assets), and whether the fund sponsor is part of a conglomerate (in %). Holdings are the share of assets invested in Treasuries and agency debt, repurchase agreements, bank deposits, bank obligations, floating-rate notes, commercial paper, and asset-backed commercial paper. Cross-sectional standard deviations of the given characteristics are in parentheses. ***, **, * represent 1%, 5%, and 10% statistical significance, respectively.

Table I: summary statistics of institutional prime money market funds.

6.5 Table II: Returns by instrument type

Read:

- Table II estimates the return contribution of different instrument holdings relative to Treasuries and agency debt.
- In the pre-period, instrument spreads are modest: risky and safe eligible instruments have similar

returns.

- In the post-period, risky instruments earn much larger spreads. Bank obligations, floating-rate notes, commercial paper, and ABCP become much more attractive in yield terms.
- The results are similar with fund fixed effects, suggesting that unobserved time-invariant fund characteristics do not drive the estimates.

Economic message: The paper's first main result is that the crisis opened up a new risk-return frontier for money funds. Risky eligible instruments suddenly paid enough to make risk-taking valuable.

TABLE II
RETURNS BY INSTRUMENT TYPE

	<i>Spread_{i,t+1}</i>			
	(1) Post	(2) Pre	(3) Post	(4) Pre
Holdings				
<i>Repurchase Agreements_{i,t}</i>	13.483 (8.638)	17.937*** (3.525)	39.748** (17.739)	11.470** (5.609)
<i>Bank Deposits_{i,t}</i>	39.445** (17.350)	17.550*** (4.044)	50.392** (20.459)	18.993*** (6.906)
<i>Bank Obligations_{i,t}</i>	88.339*** (7.929)	15.909*** (3.564)	90.868*** (19.307)	6.845 (4.885)
<i>Floating-Rate Notes_{i,t}</i>	81.953*** (7.804)	22.709*** (3.556)	89.466*** (22.629)	9.761 (6.514)
<i>Commercial Paper_t</i>	57.430*** (8.169)	16.330*** (3.370)	71.256*** (24.596)	16.252*** (5.694)
<i>Asset-backed CP_{i,t}</i>	76.619*** (8.036)	20.997*** (3.230)	84.627*** (19.852)	15.626** (6.190)
Fund characteristics				
<i>Log(Fund Size)_{i,t}</i>	0.513 (0.349)	0.259*** (0.096)	3.864** (1.946)	0.562 (0.486)
<i>Expense Ratio_{i,t}</i>	10.925*** (2.995)	1.812* (0.963)	87.327*** (25.457)	53.502*** (11.597)
<i>Age_{i,t}</i>	-1.18 (1.427)	-0.394 (0.505)	-0.576 (0.548)	-0.621 (0.457)
<i>Log(Family Size)_{i,t}</i>	-0.02 (0.225)	0.013 (0.067)	-4.352 (5.193)	0.369*** (0.139)
Constant	67.527*** (10.849)	6.031 (4.310)	68.901 (68.832)	-12.645 (8.067)
Week fixed effects	Y	Y	Y	Y
Fund fixed effects	N	N	Y	Y
Observations	7,807	11,984	7,807	11,984
R-squared	0.94	0.79	0.95	0.80

Notes. The sample is all U.S. institutional prime money market funds. The dependent variable *Spread* is computed as the annualized yield minus the Treasury bill rate. Holdings variables are the share of assets invested in repurchase agreements, bank deposits, bank obligations, floating-rate notes, commercial paper (CP), and asset-backed CP (omitted category is U.S. Treasury and agency). Fund characteristics are natural logarithm of fund size, expense ratio, fund age, and natural logarithm of fund family size. All regressions are at the weekly level and include week fixed effects. Columns (3) and (4) include fund fixed effects. Columns (1) and (3) cover the period 8/1/2007–8/31/2008 (*post* period). Columns

Table II: returns by instrument type.

6.6 Table III: Flow-performance relationship

Read:

- Table III estimates whether higher fund spreads predict subsequent inflows.
- In the post-period, the spread coefficient is positive and statistically significant.
- The economic magnitude is large: the paper reports that a one-standard-deviation increase in

fund yield raises annualized fund assets by about 43%.

- In the pre-period, the flow-performance relation is weak or absent.
- Interactions with Fund Business and Conglomerate are not economically important, suggesting that investors do not fully risk-adjust yields based on sponsor structure.

Economic message: The table provides the incentive mechanism. Institutional investors chase yield, so sponsors can attract assets by increasing fund yield, even when higher yield reflects higher risk.

TABLE III
FLOW-PERFORMANCE RELATIONSHIP

	<i>Fund Flow</i> _{<i>i,t+1</i>}					
	(1) Post	(2) Pre	(3) Post	(4) Pre	(5) Post	(6) Pre
<i>Spread</i> _{<i>i,t</i>}	0.013*** (0.005)	0.003 (0.005)	0.024*** (0.008)	0.000 (0.004)	0.026*** (0.009)	0.009 (0.010)
<i>Fund Business</i> _{<i>i,2006</i>} *					-0.003 (0.006)	-0.009 (0.009)
<i>Spread</i> _{<i>i,t</i>}					0.000 (0.003)	0.003 (0.005)
<i>Conglomerate</i> _{<i>i,2006</i>} *						
<i>Spread</i> _{<i>i,t</i>}						
<i>Log(Fund Size)</i> _{<i>i,t</i>}	-0.120** (0.051)	-0.077*** (0.029)	-7.658*** (1.341)	-4.146*** (0.720)	-7.655*** (1.345)	-4.148*** (0.720)
<i>Expense Ratio</i> _{<i>i,t</i>}	-0.549* (0.320)	-1.276*** (0.354)	-2.668 (5.965)	-1.365 (3.703)	-2.680 (5.915)	-1.475 (3.704)
<i>Age</i> _{<i>i,t</i>}	0.159 (0.180)	-0.078 (0.149)	0.015 (0.322)	0.715** (0.323)	0.014 (0.322)	0.713** (0.323)
<i>Flow Volatility</i> _{<i>i,t</i>}	4.245* (2.322)	2.476** (1.243)	1.381 (3.175)	-0.213 (2.152)	1.331 (3.166)	-0.230 (2.146)
<i>Log(Family Size)</i> _{<i>i,t</i>}	0.026 (0.023)	0.032** (0.014)	0.528 (1.238)	0.042 (0.126)	0.521 (1.245)	0.045 (0.127)
Week fixed effects	Y	Y	Y	Y	Y	Y
Fund fixed effects	N	N	Y	Y	Y	Y
Observations	7,807	11,984	7,807	11,984	7,807	11,984
<i>R</i> -squared	0.022	0.017	0.085	0.052	0.085	0.052

Notes. The sample is all U.S. institutional prime money market funds. Columns (1), (3), and (5) cover the period from 8/1/2007 to 8/31/2008 (*post* period). Columns (2), (4), and (6) cover the period from 1/1/2006 to 7/31/2007 (*pre* period). The dependent variable is *Fund Flow*, computed as the percentage change in total net assets from time t to time $t+1$, adjusted for market appreciation and winsorized at the 0.5% level. Independent variables are the weekly annualized spread from t to $t-1$, natural logarithm of fund size, fund expense ratio, fund age, volatility of fund flows based on past 12-week fund flows, and natural logarithm of fund family size. In columns (5) and (6), additional independent variables are the interactions of *Spread* with *Fund Business* and *Conglomerate*. *Fund Business* (FB) is mutual fund assets other than institutional prime fund assets as a share of total sponsor's mutual fund assets. *Conglomerate* is an indicator variable equal to 1 if the fund sponsor is affiliated with a financial conglomerate and 0 otherwise. All regressions are at the weekly level and include week fixed effects. Columns (3) to (6) also include fund fixed effects. Standard errors are clustered at the fund level. ***, **, * represent 1%, 5%, and 10% statistical significance, respectively.

Table III: flow-performance relationship.

6.7 Figure V and Table IV: Sponsor business spillovers and risk taking

Read:

- Figure V plots the time-varying relation between Fund Business and three risk measures: holdings risk, maturity risk, and spread.
- Before August 2007, the relation is relatively stable and does not show strong risk differentiation.
- After August 2007, the coefficient becomes sharply negative: funds with more non-money-fund business take less risk.
- Table IV confirms this pattern in regressions with week fixed effects, sponsor fixed effects, and fund fixed effects.
- The negative post interaction means that sponsors with more franchise value outside institutional prime funds reduce risk relative to more specialized money-fund sponsors.

Economic message: This is the core cross-sectional result. The same shock creates different behavior depending on sponsor incentives. Sponsors with more outside business protect reputation; specialized sponsors chase yield more aggressively.

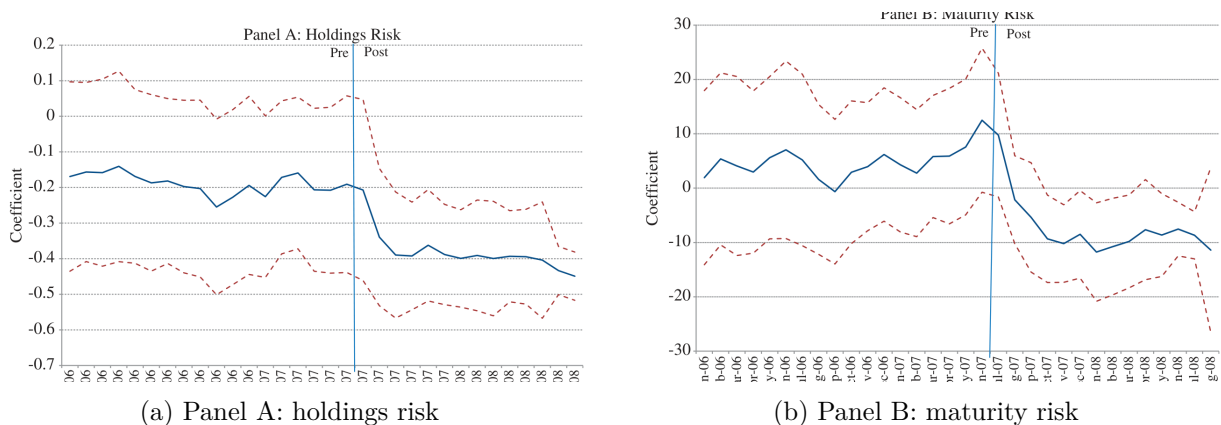


Figure V, Panels A and B: sponsor fund business spillovers and risk taking.

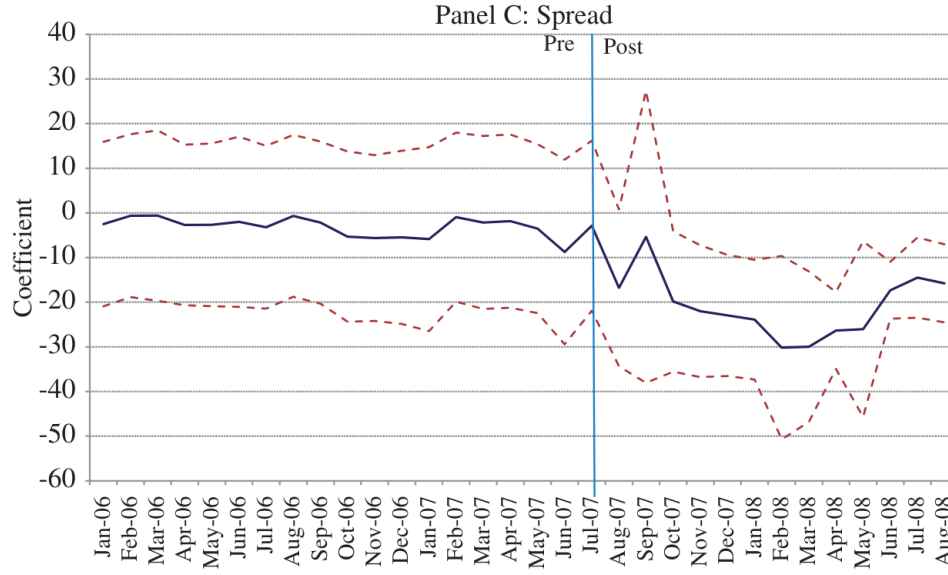


FIGURE V

Figure V, Panel C: spread.

TABLE IV
SPONSOR'S BUSINESS SPILLOVERS AND RISK TAKING

	<i>Holdings Risk_{i,t+1}</i>			<i>Maturity Risk_{i,t+1}</i>			<i>Spread_{i,t+1}</i>		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Fund Business_{i,2006}</i> * <i>Post_t</i>	-18.346* (9.298)	-21.289** (8.705)	-19.636** (8.942)	-11.449** (5.335)	-12.588** (5.782)	-11.902** (5.910)	-15.040** (7.470)	-15.473** (7.443)	-14.218* (7.419)
<i>Conglomerate_{i,2006}</i> * <i>Post_t</i>	-6.781** (3.017)	-5.580* (2.885)	-6.326** (2.962)	-1.665 (1.722)	-1.520 (1.750)	-1.703 (1.762)	-7.263*** (2.429)	-7.215*** (2.424)	-7.321*** (2.428)
<i>Fund Business_{i,2006}</i>	-18.126 (13.540)			5.398 (5.310)			-2.764 (1.929)		
<i>Conglomerate_{i,2006}</i>	-6.539 (4.233)			-1.698 (1.938)			-1.212* (0.655)		
<i>Controls_{i,2006}</i>	Y	Y	Y	Y	Y	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sponsor fixed effects	N	Y	N	N	Y	N	N	Y	N
Fund fixed effects	N	N	Y	N	N	Y	N	N	Y
Observations	19,096	19,096	19,096	19,096	19,096	19,096	19,096	19,096	19,096
R-squared	0.209	0.624	0.780	0.142	0.482	0.587	0.952	0.957	0.959

Notes. The sample is all U.S. institutional prime money market funds for the period from 1/1/2006 to 8/31/2008. The dependent variables are: the fraction of assets held in risky assets net of the riskless assets (*Holdings Risk*) in columns (1)–(3), average portfolio maturity (*Maturity Risk*) in columns (4)–(6), and the weekly annualized spread (*Spread*) in columns (7)–(9). *Fund Business* is the sponsor's share of mutual fund assets other than institutional prime money market funds in total sponsor's assets. *Conglomerate* is an indicator variable equal to 1 if the fund sponsor is affiliated with a financial conglomerate, and 0 otherwise. *Post* is an indicator variable equal to 1 for the period from 8/1/2007 to 8/31/2008, and 0 otherwise. The other independent variables (*Controls*) are fund assets, expense ratio, fund age, and fund family size as of 1/1/2006, and interactions of these variables with *Post* (coefficients not shown). All regressions are at the weekly level and include week fixed effects. Columns (2), (5), and (8) include sponsor fixed effects, and columns (3), (6), and (9) include fund fixed effects. Standard errors are clustered at the sponsor level. ***, **, * represent 1%, 5%, and 10% statistical significance, respectively.

Table IV: sponsor business spillovers and risk taking.

6.8 Table V: Evidence from retail funds

Read:

- Table V repeats the flow-performance and risk-taking tests for retail prime funds.
- Retail investors are less responsive to yield in the post-period than institutional investors.
- The business-spillover variables do not predict retail fund risk taking in the same way.
- This placebo result helps rule out a pure sponsor-risk-culture explanation.

Economic message: The retail evidence supports the mechanism: risk taking is strongest where investors chase yield most aggressively. The same sponsor does not show the same pattern where the investor base is less yield sensitive.

TABLE V
EVIDENCE FROM RETAIL FUNDS

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Fund Flow_{i,t+1}</i>					
	Post	Pre	Post	Pre	Post	Pre
Panel A: Flow–performance relationship						
<i>Spread_{i,t}</i>	0.002 (0.002)	0.008** (0.003)	0.003 (0.003)	0.008** (0.003)	0.008 (0.005)	0.021*** (0.005)
<i>Retail Fund</i>					−0.007 (0.005)	−0.015** (0.007)
<i>Business_{i,2006}*</i>						
<i>Spread_{i,t}</i>						
<i>Conglomerate_{i,2006}*</i>					−0.001 (0.002)	0.002 (0.004)
<i>Spread_{i,t}</i>						
<i>Controls_{i,t}</i>	Y	Y	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y	Y	Y
Fund fixed effects	N	N	Y	Y	Y	Y
Observations	3,724	6,004	3,724	6,004	3,724	6,004
R-squared	0.058	0.022	0.109	0.075	0.110	0.076
Panel B: Business spillovers and risk taking						
	<i>Holdings Risk_{i,t+1}</i>		<i>Maturity Risk_{i,t+1}</i>		<i>Spread_{i,t+1}</i>	
<i>Retail Fund</i>	16.098 (14.417)	15.229 (14.332)	2.491 (6.324)	2.263 (6.276)	−8.427 (14.702)	−8.076 (14.803)
<i>Business_{i,2006}*</i>						
<i>Post_t</i>						
<i>Conglomerate_{i,2006}*</i>	7.072 (6.125)	6.726 (6.178)	−4.045 (2.529)	−3.989 (2.643)	−4.017 (5.455)	−3.973 (5.466)
<i>Post_t</i>						
<i>Controls_{i,2006}</i>	Y	Y	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y	Y	Y
Sponsor fixed effects	Y	N	Y	N	Y	N
Fund fixed effects	N	Y	N	Y	N	Y
Observations	9,492	9,492	9,492	9,492	9,492	9,492
R-squared	0.772	0.74	0.606	0.631	0.909	0.911

Notes. The sample is all U.S. retail prime money market funds for the period from 1/1/2006 to 8/31/2008. *Business_{i,2006}** is the business risk score for the fund's top 10 holdings in 2006. *Conglomerate_{i,2006}** is the conglomerate risk score for the fund's top 10 holdings in 2006. *Post_t* is a dummy variable equal to 1 if the fund is in the post-period (after 8/31/2008) and 0 otherwise.

Table V: evidence from retail funds.

6.9 Table VI: Risk taking after government guarantee

Read:

- Table VI extends the risk-taking regression to include the period after the government guarantee.
- The business-spillover effect is strong in the August 2007 to August 2008 post period.
- The effect largely disappears in the post-Lehman period after explicit government guarantees are introduced.
- This weakens the idea that the main results are driven by pre-existing expectations of government bailouts.

Economic message: If everyone is explicitly insured, sponsor franchise value matters less for risk choices. This supports the interpretation that business spillovers, not just bailout expectations, discipline risk taking before Lehman.

TABLE VI
RISK TAKING AFTER GOVERNMENT GUARANTEE

	<i>Holdings Risk_{i,t+1}</i>		<i>Maturity Risk_{i,t+1}</i>		<i>Spread_{i,t+1}</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Fund Business_{i,2006}*Post_t</i>	-22.319** (8.529)	-19.636** (8.935)	-12.465** (5.625)	-11.902** (5.906)	-15.326** (7.429)	-14.218* (7.413)
<i>Fund Business_{i,2006}*Post-Lehman_t</i>	-22.999 (18.611)	-9.592 (19.229)	-9.419 (6.488)	-5.162 (6.283)	-2.965 (7.175)	-1.057 (7.502)
<i>Conglomerate_{i,2006}*Post_t</i>	-5.278* (2.899)	-6.326** (2.959)	-1.399 (1.728)	-1.703 (1.760)	-7.123*** (2.413)	-7.321*** (2.426)
<i>Conglomerate_{i,2006}*Post-Lehman_t</i>	-0.860 (5.894)	-3.569 (5.649)	-4.081* (2.159)	-5.010** (2.131)	-3.441 (2.168)	-4.044* (2.237)
<i>Controls_{i,2006}</i>	Y	Y	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y	Y	Y
Sponsor fixed effects	Y	N	Y	N	Y	N
Fund fixed effects	N	Y	N	Y	N	Y
Observations	28,409	28,409	28,409	28,409	28,409	28,409
R-squared	0.579	0.694	0.457	0.537	0.953	0.954

Notes. The sample is all U.S. institutional prime money market funds for the period from 1/1/2006 to 12/31/2010. We estimate the same regression models as in Table IV for the period from July 2006 to December 2010. We drop the month of the Lehman's bankruptcy and the two quarters immediately after the Lehman's bankruptcy to focus on risk taking after a short adjustment period. We interact our main variables of interest with an indicator variable for the post period (July 2007–August 2008) and the post-Lehman period (April 2009–December 2010). All regressions include the control variables specified in Table IV and interactions of the control variables with the post-Lehman indicator variable (coefficients not shown). The regressions are at the weekly level and include week fixed effects. Columns (1), (3), and (5) include sponsor fixed effects, and columns (2), (4), and (6) include fund fixed effects. Standard errors are clustered at the sponsor level. ***, **, * represent 1%, 5%, and 10% statistical significance, respectively.

Table VI: risk taking after government guarantee.

6.10 Table VII: Controlling for financial strength

Read:

- Table VII focuses on funds sponsored by stand-alone asset managers.
- This narrows the sample to sponsors that are more similar in their ability to support funds during stress.
- Fund Business remains negatively related to risk taking after August 2007.

- Controls for whether the sponsor has no credit rating do not eliminate the main effect.

Economic message: The main result is not only about ability to bail out a fund. Even among stand-alone sponsors, the amount of non-money-fund business affects risk-taking incentives.

TABLE VII
CONTROLLING FOR FINANCIAL STRENGTH

	Stand-alone asset managers					
	<i>Holdings Risk_{i,t+1}</i>		<i>Maturity Risk_{i,t+1}</i>		<i>Spread_{i,t+1}</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
Fund <i>Business_{i,2006}</i> *	-39.968***	-38.720***	-11.210**	-10.029**	-22.829***	-22.413***
<i>Post_t</i>	(12.720)	(12.656)	(4.462)	(4.403)	(7.399)	(7.470)
<i>No Rating_{i,2006}</i> *	-8.901	-8.273	-2.121	-1.689	-7.858**	-7.777**
<i>Post_t</i>	(5.394)	(5.406)	(1.437)	(1.355)	(3.082)	(3.065)
<i>Controls_{i,2006}</i>	Y	Y	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y	Y	Y
Sponsor fixed effects	Y	N	Y	N	Y	N
Fund fixed effects	N	Y	N	Y	N	Y
Observations	7,645	7,645	7,645	7,645	7,645	7,645
R-squared	0.607	0.715	0.589	0.671	0.968	0.968

Notes. The sample is all U.S. institutional prime money market funds affiliated with an independent asset manager for the period 1/1/2006–8/31/2008. The dependent variables, *Fund Business* and *Post*, are defined in Table IV. *No Rating* is an indicator variable equal to 1 if the sponsor has no credit rating, and 0 otherwise. All regressions include the same control variables as in Table IV (coefficients not shown). They are at the weekly level and include week fixed effects and fund fixed effects. Standard errors are clustered at the sponsor level. ***, **, * represent 1%, 5%, and 10% statistical significance, respectively.

Table VII: controlling for financial strength.

6.11 Table VIII: Post-Lehman results

Read:

- Table VIII examines redemptions, sponsor support, exit, and fund name changes after Lehman’s default.
- Funds with higher pre-run spreads experience larger redemptions.
- Greater Fund Business predicts lower redemptions, consistent with investors trusting sponsors with more franchise value at stake.
- Conglomerate sponsors are more likely to provide support.
- Funds with greater business-spillover concerns are also more likely to adjust through exit or name changes after the crisis.

Economic message: The ex post results complete the mechanism. Risk taking creates run exposure, and sponsor reputation matters when investors decide whether to redeem and when sponsors decide whether to support or restructure funds.

TABLE VIII
POST-LEHMAN RESULTS

	(1) Redemptions	(2) Support	(3) Exit	(4) Name change
<i>Fund Business_i</i>	−0.159** (0.074)	−0.119 (0.366)	−0.156 (0.247)	0.334* (0.174)
<i>Conglomerate_i</i>	0.001 (0.029)	0.294** (0.133)	0.243*** (0.079)	−0.018 (0.101)
<i>Spread_i</i>	0.001** (0.000)	0.001 (0.002)	0.001 (0.001)	0.000 (0.002)
<i>Log(Fund Size)_i</i>	0.026*** (0.007)	0.019 (0.015)	−0.061** (0.030)	0.023 (0.018)
<i>Expense Ratio_i</i>	−0.191*** (0.055)	−0.019 (0.185)	−0.188 (0.207)	−0.216 (0.504)
<i>Age_i</i>	−0.028 (0.033)	−0.086 (0.083)	0.07 (0.059)	0.012 (0.067)
<i>Log(Family Size)_i</i>	0.015*** (0.004)	0.03 (0.022)	0.028* (0.016)	−0.002 (0.007)
Constant	−0.041 (0.193)	0.273 (0.444)	0.166 (0.360)	−0.369 (0.389)
Observations	105	105	105	105
<i>R</i> -squared	0.381	0.189	0.135	0.053

Notes. The sample is all U.S. institutional prime money market funds that were active from 1/1/2006 until 10/1/2009. In column (1) the dependent variable is *Redemptions* defined as total value of redemptions (fund outflows) in the week after Lehman's bankruptcy (9/18–25/2008). In column (2) the dependent variable is *Support*, an indicator variable equal to 1 if the fund's sponsor filed a no-action letter with the SEC in the week after the Lehman's bankruptcy and 0 otherwise (20 funds declared support). In column (3) the dependent variable is *Exit*, an indicator variable equal to 1 if the fund was closed in the two years after the expiration of the government guarantee (10/1/2009), and 0 otherwise (16 out of 105 fund closures). In column (4) the dependent variable is *Name*, an indicator variable equal to 1 if the fund name was changed to match the sponsor name, 0 if the name was unchanged, and equal to −1 if the fund name was changed to be different from the sponsor name (eight name changes). All independent variables are defined in Table III. In columns (1) to (3), the independent variables are measured as of the week before the Lehman bankruptcy. In column (4), the independent variables are defined as of the end of the

Table VIII: post-Lehman results.

6.12 Appendix Table A.1: Post-Lehman support arrangements

Read:

- Appendix Table A.1 lists support arrangements after Lehman's bankruptcy.
- Support events involve many sponsors, including BNY Mellon, USAA, Touchstone, SunTrust, Principal, Morgan Stanley, Bank of America, ING, Legg Mason, and others.
- Support often takes the form of capital support agreements or cash contributions intended to maintain fund value near the stable NAV threshold.

- Distressed assets include Lehman notes, AIG notes, Morgan Stanley-related exposures, and other eligible assets.

Economic message: Sponsor support was not a purely theoretical possibility. During the run, sponsors faced real decisions about whether and how to protect their funds and reputations.

Fund company	Sponsor	Support date	Distress reason	Value of distressed assets	Support value	Remarks
Dreyfus Cash Mgmt. Plus	BNY Mellon	10/29/2008	Lehman Brothers notes	\$97.1M		Cash contribution necessary to maintain the fund value at 0.995
All Dreyfus funds	BNY Mellon	10/29/2008	Distress of eligible assets			CSA (Cash contribution necessary to maintain the fund value at 0.995)
All Citizens funds	BNY Mellon	10/29/2008	Distress of eligible assets			CSA (Cash contribution necessary to maintain the fund value at 0.995)
All General funds	BNY Mellon	10/29/2008	Distress of eligible assets			CSA (Cash contribution necessary to maintain the fund value at 0.995)
Dreyfus Basic MMF	BNY Mellon	10/29/2008	Lehman Brothers notes	\$45M		Cash contribution necessary to maintain the fund value at 0.995
Dreyfus LAP	BNY Mellon	10/29/2008	Lehman Brothers notes	\$190M		Cash contribution necessary to maintain the fund value at 0.995
Dreyfus Worldwide Dollar MMF	BNY Mellon	10/29/2008	Lehman Brothers notes	\$20M		Cash contribution necessary to maintain the fund value at 0.995
Russell MMF	Northwestern Mutual Life Ins.	10/20/2008	The entire fund			CSA (Cash contribution necessary to maintain the fund value at 0.995)
USAA MMF	USAA	10/22/2008	AIG notes	\$81.96M		CSA (Cash contribution necessary to maintain the fund value at 0.995)
	Touchstone Advisors	10/22/2008				

Fund company	Sponsor	Support date	Distress reason	Value of distressed assets	Support value	Remarks
Touchstone Invest. Trust Instit. MMF			Morgan Stanley, Southtrust Bank, Wachovia notes	MS (\$5.06M), ST (\$1.4M), Wach. (\$8.35M)		Cash contribution necessary to maintain the fund value at 0.995 (LOC by Western and Southern Life Insurance Company)
Touchstone Invest. Trust MMF	Touchstone Advisors	10/22/2008	Morgan Stanley, Southtrust Bank, Wachovia notes	MS (\$5.1M), ST (\$1.4M), Wach. (\$4.47M)		Cash contribution necessary to maintain the fund value at 0.995 (LOC by Western and Southern Life Insurance Company)
Touchstone Variable Series MMF	Touchstone Advisors	10/22/2008	Morgan Stanley, Southtrust Bank, Wachovia notes	MS (\$2.25M), Wach. (\$1.5M)		Cash contribution necessary to maintain the fund value at 0.995 (LOC by Western and Southern Life Insurance Company)
Tamarack Prime MMF	Voyageur Asset Management	10/22/2008	The entire fund			Cash contribution necessary to maintain the fund value at 0.995 (LOC by RBC)
Tamarack Instit. Prime MMF	Voyageur Asset Management	10/22/2008	The entire fund			Cash contribution necessary to maintain the fund value at 0.995 (LOC by RBC)
RidgeWorth Prime Quality MMF	Lehman Brothers notes	10/22/2008	Lehman Brothers notes	\$70M	\$70M	Exchange of FirstTrust Note for the Lehman note in the amount of \$70M
Principal MMF	Principal Financial Group	10/22/2008	AIG notes			CSA (Cash contribution necessary to maintain the fund value at 0.995)
Principal Variable Contracts MMF	Principal Financial Group	10/22/2008	AIG notes			CSA (Cash contribution necessary to maintain the fund value at 0.995)
Morgan Stanley funds	Morgan Stanley	10/22/2008	The entire fund			CSA (Cash contribution necessary to maintain the fund value at 0.995)
Active Assets funds	Morgan Stanley	10/22/2008	The entire fund			CSA (Cash contribution necessary to maintain the fund value at 0.995)
Columbia MM Reserves	Bank of America	10/22/2008	The entire fund			CSA (Cash contribution necessary to maintain the fund value at 0.995)

(a) Part 1

(b) Part 2

Appendix Table A.1: support arrangements, parts 1 and 2.

Fund company	Sponsor	Support date	Distress reason	Value of distressed assets	Support value	Remarks
ING LAP	ING Groep N.V.	10/22/2008	AIG notes	\$46M		CSA (Cash contribution necessary to maintain the fund value at 0.995)
ING Classic MMF	ING Groep N.V.	10/22/2008	AIG notes	\$28M		CSA (Cash contribution necessary to maintain the fund value at 0.995)
ING Instit. Prime MMF	ING Groep N.V.	10/22/2008	AIG notes	\$46M		CSA (Cash contribution necessary to maintain the fund value at 0.995)
ING MMF	ING Groep N.V.	10/22/2008	AIG notes; Lehman notes	AIG (\$8.5M), Lehman (\$2M)		CSA (Cash contribution necessary to maintain the fund value at 0.995)
ING Brokerage Cash Reserves	ING Groep N.V.	10/22/2008	AIG notes	\$8M		CSA (Cash contribution necessary to maintain the fund value at 0.995)
Western Asset Instit. MMF	Legg Mason	10/22/2008	Orion Finance. LLC notes (SIV)	\$75M	\$20M	CSA
Western Asset Instit. MMF	Legg Mason	10/22/2008	The fraction of fund	\$452M		CSA (Cash contribution necessary to maintain the fund value at 0.9975)
Russell MMF	Northwestern Mutual Life	10/24/2008	Lehman Brothers notes	\$403M		CSA (Cash contribution necessary to maintain the fund value at 0.995)

Notes. The columns include the fund company offered support, the sponsor company providing support, the support date, the reason for support, the values of distressed securities, the value of support, and additional remarks. CSA: Capital Support Agreement; POF: Prime Obligations Fund; DAP: Diversified Assets Portfolio; LAP: Liquid Assets Portfolio; LRP: Liquid Reserves Portfolio; POP: Prime Obligations Portfolio.

Appendix Table A.1: support arrangements, part 3.

7 Short summary

- The paper studies risk taking by money market funds during the financial crisis of 2007-2010.
- Money market funds looked safe because of stable NAV accounting, high-rated short-term assets, and a cash-like investor experience.
- After August 2007, risky eligible instruments began paying much higher spreads than safe government-related instruments.
- Institutional investors chased yield, so higher-yield funds attracted large inflows.

- This created incentives for sponsors to take risk, because sponsor revenue increases with assets under management.
- Sponsors with more non-money-fund business took less risk, consistent with larger reputational spillovers if a money fund failed.
- Stand-alone asset managers and sponsors with more money fund business took more risk.
- Retail fund evidence, post-guarantee evidence, and financial-strength controls support the business-spillover interpretation.
- Risk taking mattered during the run: higher-spread funds suffered larger redemptions after Lehman's default.
- Sponsor support, fund exit, and name changes after Lehman further show that franchise value shaped crisis behavior.

8 Conclusion

8.1 Core conclusion

Kacperczyk and Schnabl show that money market funds are not inherently safe simply because they hold short-term, highly rated assets and maintain a stable NAV. When risk premia open up, funds can have strong incentives to chase yield. Institutional investors reward higher current yield with inflows, and fund sponsors earn fees on those assets. As a result, money funds can take on risk precisely when the financial system is fragile.

8.2 Economic intuition

The central intuition is a trade-off between fee income and franchise-value protection. A fund can raise yield by investing in riskier instruments. Higher yield attracts institutional flows, which increases sponsor fee income. But if the risky assets deteriorate, the fund may face redemptions or break the buck, which can damage the sponsor's broader business. Sponsors with more non-money-fund business internalize more of this reputational cost and therefore take less risk. Sponsors specialized in money funds have less outside business to protect and therefore have stronger incentives to chase yield.

8.3 Incremental contribution revisited

The paper contributes by combining a sharp crisis-period change in risk-taking opportunities, high-frequency fund-level data, detailed portfolio information, and sponsor-level business structure. It moves the literature from asking whether money market funds experienced runs to explaining why some funds had stronger incentives to take the risks that made runs possible.

8.4 Implications for financial stability and regulation

The findings imply that stable NAV accounting and asset-quality restrictions do not fully eliminate money fund risk. If eligible assets can become risky and investors chase small yield differences, fund sponsors may reach for yield. Regulation therefore needs to address incentives, run vulnerability, and sponsor support expectations, not just nominal asset eligibility.

8.5 Limitations

The paper's design is observational and cannot rule out every unobserved time-varying sponsor characteristic. Some risk is measured through broad instrument categories rather than complete security-level credit exposures. The setting is especially relevant for institutional prime funds during the crisis, so one should be cautious in applying the same magnitudes to retail funds, Treasury funds, or later regulatory regimes.

8.6 Takeaway

The enduring takeaway is that money market funds are safe only when their incentives and asset opportunities make them safe. During the crisis, those conditions failed: risky money market instruments paid much higher yields, investors chased those yields, some sponsors took risk, and the sector became vulnerable to runs. The paper turns the question "How safe are money market funds?" into an incentive-based answer: they are as safe as the risk-taking incentives of their sponsors and investors allow them to be.